Final Presentation
T3 - Autonomous Driving

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Task Definition

Definition: What Problem is Being Addressed?

3D bounding-box detection for partially occluded vehicles via monocular lenses mounted on self-driving vehicles.

Constraints

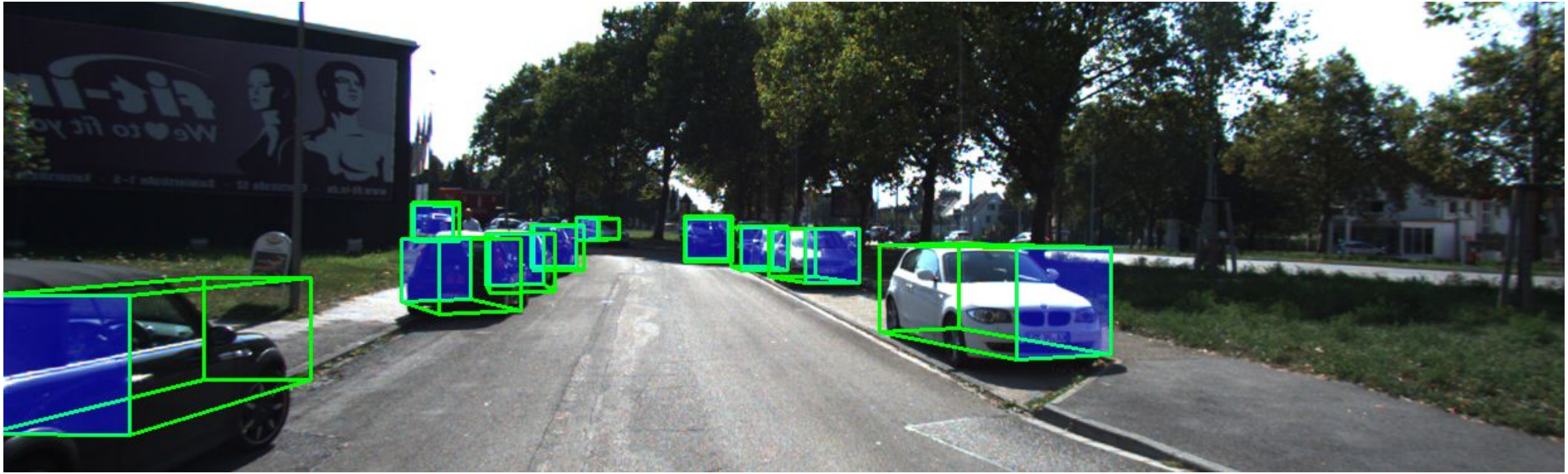
- Single lens / Monocular
- Stretch Goal: ≤ 50ms classification time

Task Definition

Input / Output: KITTI Dataset

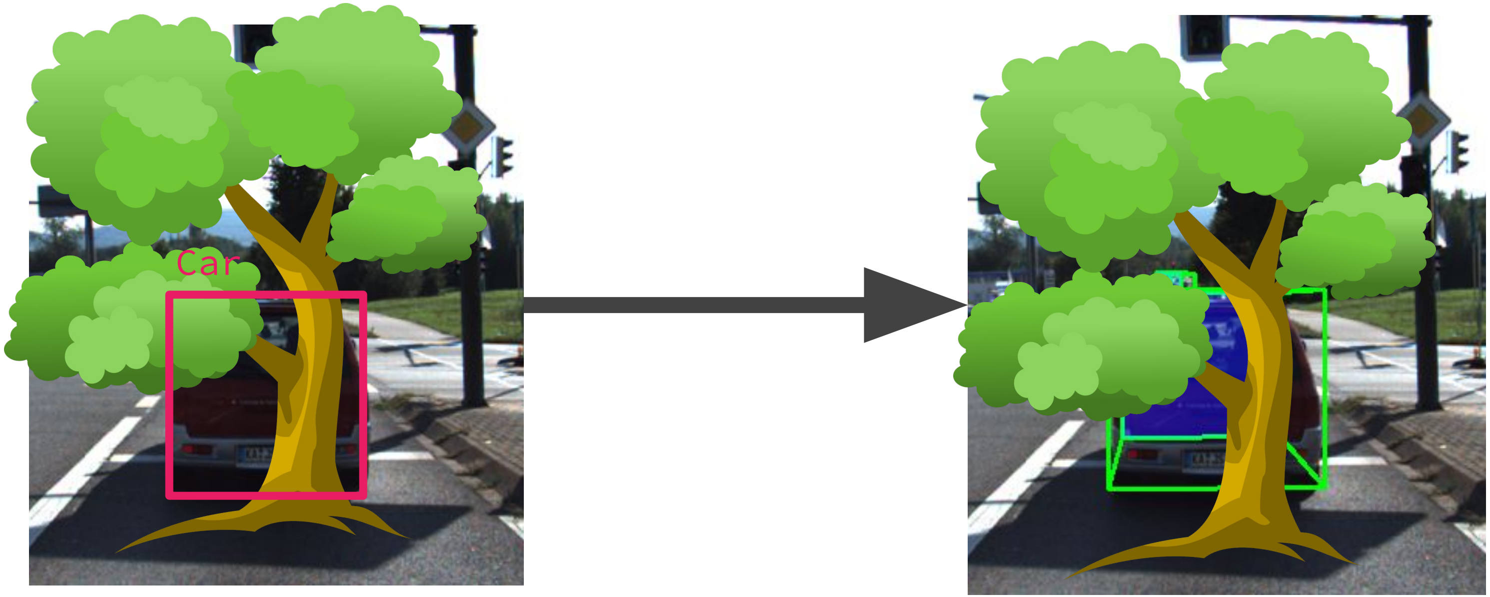
Task Definition

Input / Output: KITTI Dataset



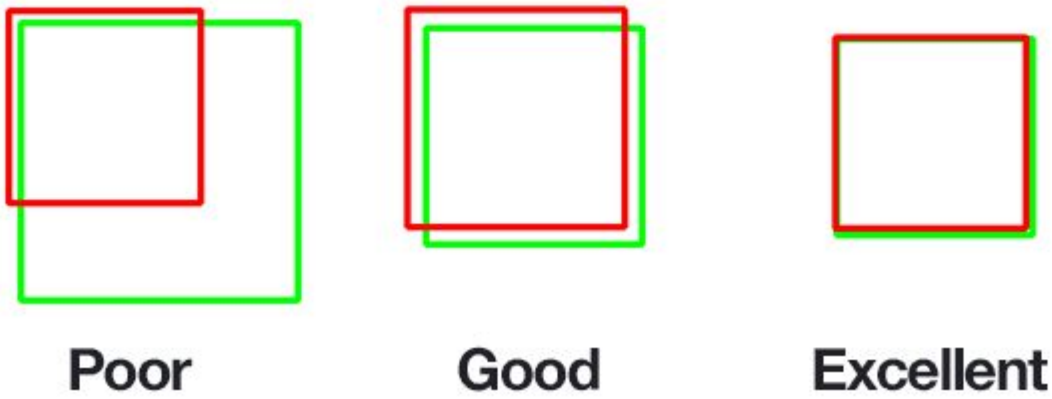
Task Definition

Simple Example



Evaluation: Summary

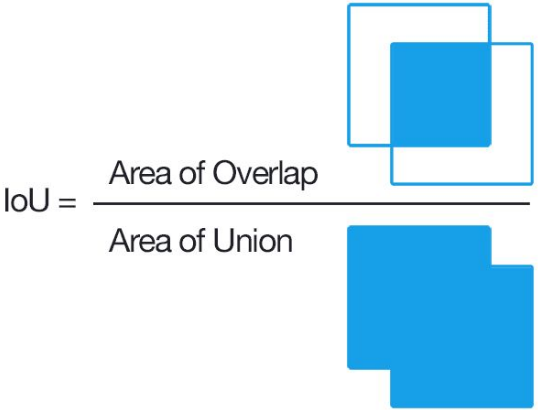
- 3D Rotated IOU
- Average Precision

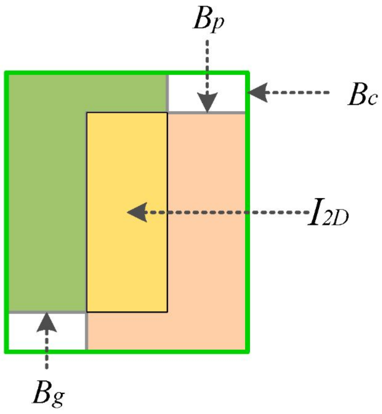


Task Definition

Evaluation: 3D Rotated IOU

- 3D Rotated IOU
- Average Precision



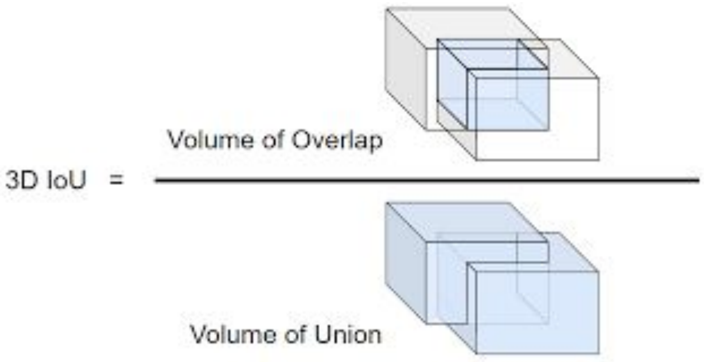


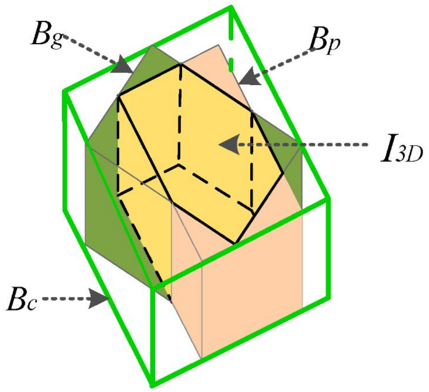
2D IOU

Task Definition

Evaluation: 3D Rotated IOU

- 3D Rotated IOU
- Average Precision





3D Rotated IOU

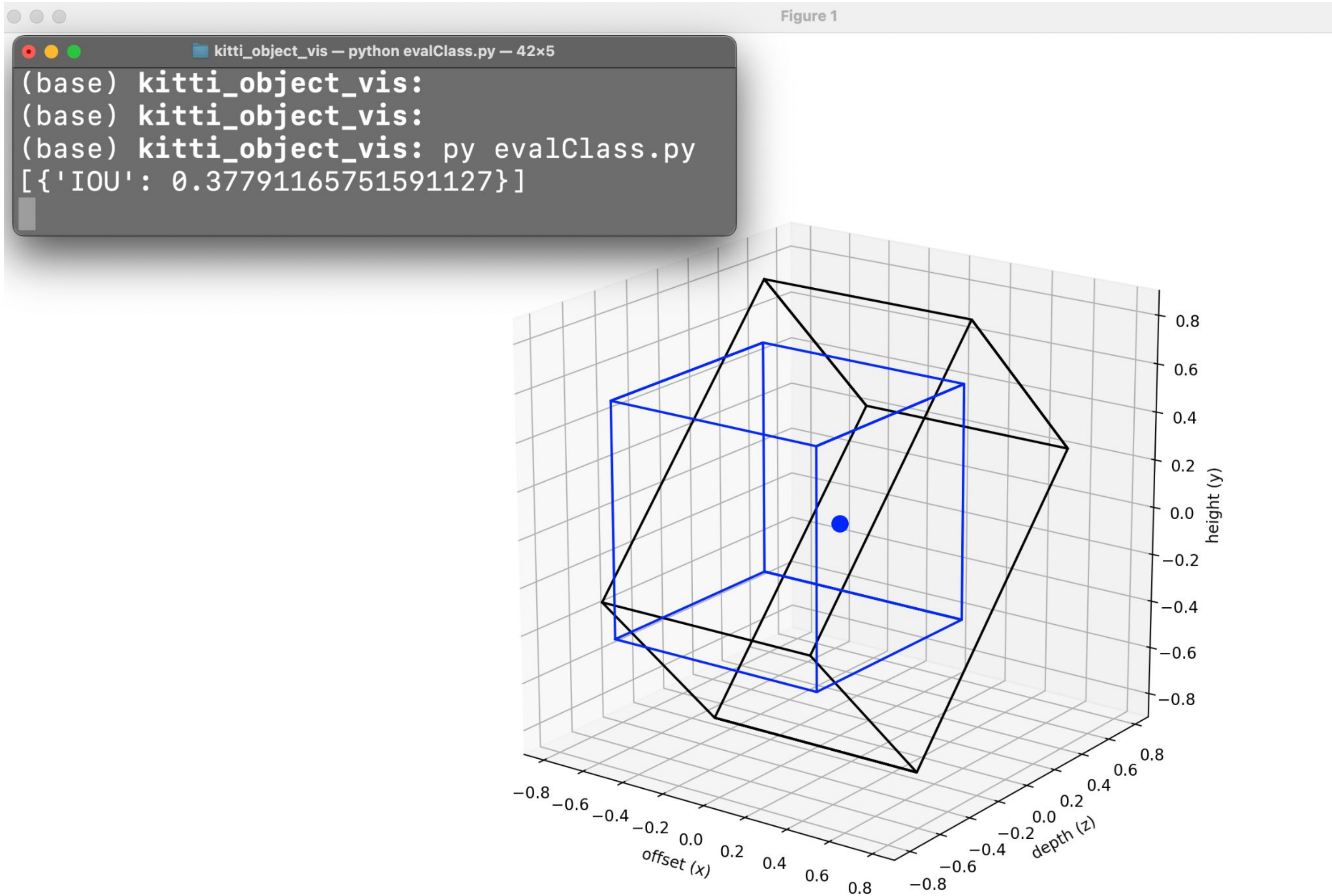


Figure 1
kitti_object_vis — python evalClass.py — 42×5
(base) kitti_object_vis:
(base) kitti_object_vis:
(base) kitti_object_vis: py evalClass.py
[{'IOU': 0.37791165751591127}]
height (y)
offset (x)
depth (z)
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8

Evaluation: Average Precision

- 3D Rotated IOU
- Average Precision

1. **Choose IOU Threshold**

Recall = IOU

2. **Calculate Precision**

Precision = TP / (TP + FP)

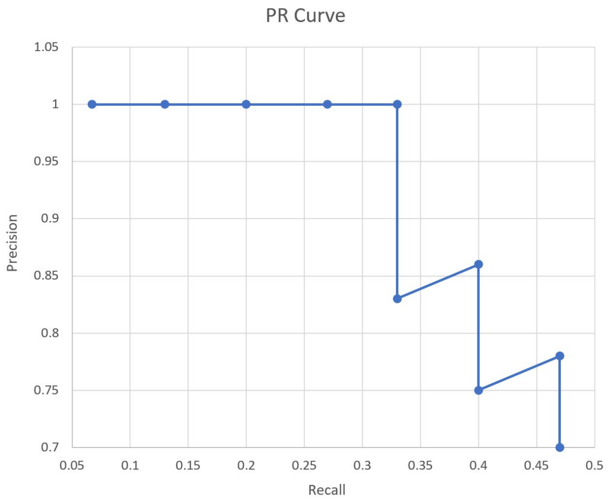
Number of bounding boxes processed so far

Task Definition

Evaluation: Average Precision

- 3D Rotated IOU
- Average Precision

4. Calculate Area Under Precision-Recall Curve



This area is the AP

larger AP => better model

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Motivation

Importance: Why is This Task Important?

- Occlusion is extremely common in driving scenarios.

- Good decision-making requires seeing as many objects as possible
  - **Including partially occluded objects**
  - Why not just 2D?

Motivation

Non-Trivial: Why isn't This Task Trivial?

---

1. Estimating depth in an rgb image

2. Many types of occlusion

Non-Trivial: Why isn't This Task Trivial?

---

1. **Estimating depth in an rgb image**

2. Many types of occlusion

Motivation

How wide is this car?

How wide is this car?

75.5 in

=

1.9177 meters

What about this one?

0 meters (it's a billboard!)

Motivation

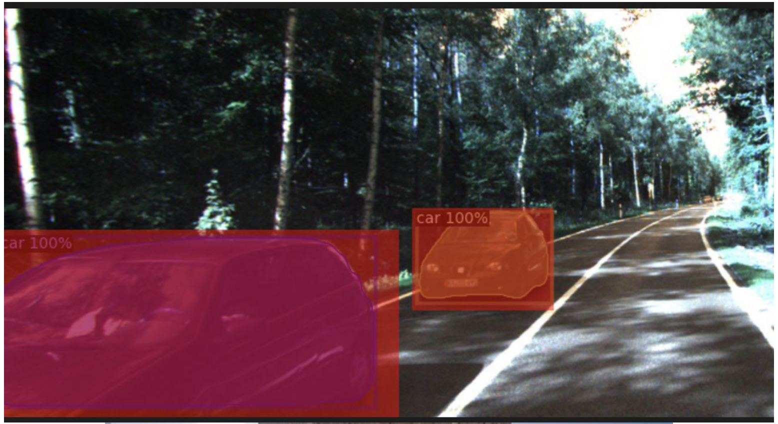
Non-Trivial: Why isn't This Task Trivial?

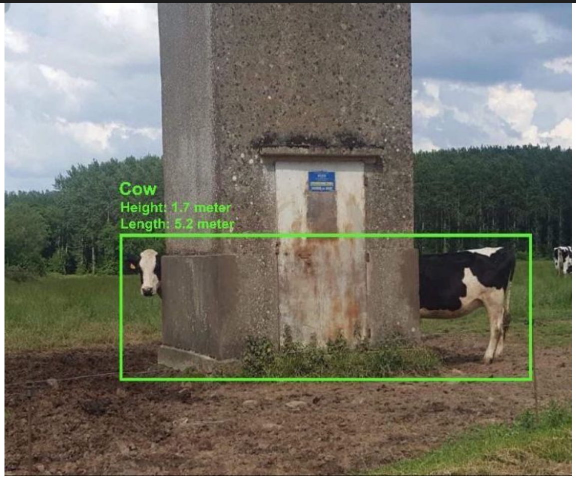
---

1. Estimating depth in an RGB image

2. **Many types of occlusion**

Motivation





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Others' Approaches: YOLO3D

1.

Uses a YOLO model to first identify objects in the image.

- YOLO outputs a 2D bounding box + object category

Approaches

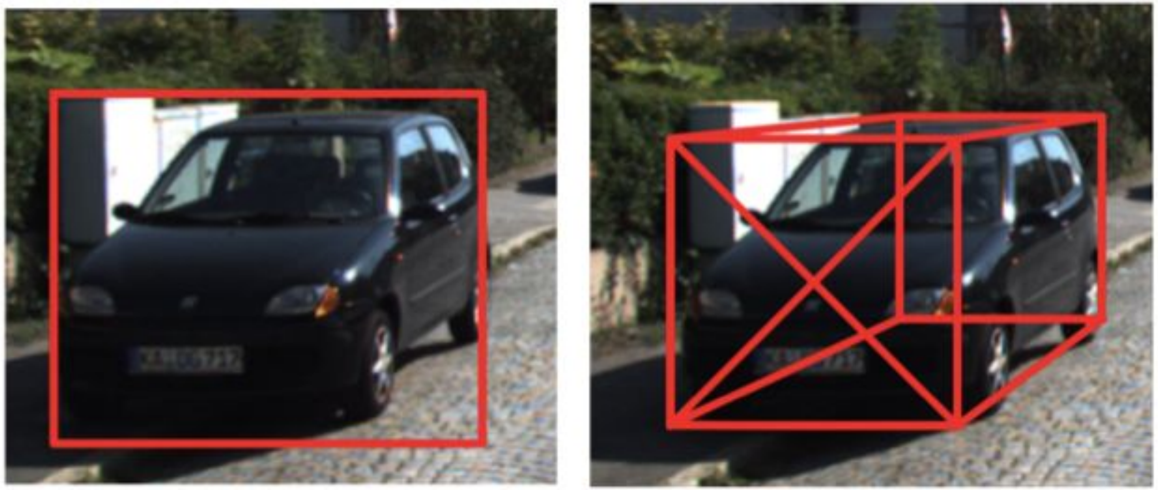
Others' Approaches: YOLO3D

2.

Take the output 2D box and, using another model (CNN), regress the 3D object properties (orientation and dimensions)

Approaches

Others' Approaches: YOLO3D



3.

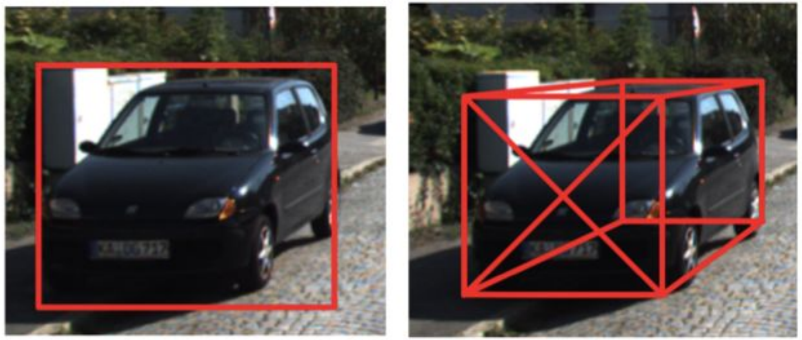
Finally, combine the 2D bounding box, the regressed object orientation and dimensions, and apply geometric constraints to estimate the 3D bounding box

Approaches

Others’ Approaches: YOLO3D

Image frame = 2D bounding box

Circles represent points that are active constraints



result

Our Approach: How Did We Address This Task?

- Didn't end up using multiple images -
  - It increases classification time to be slower than binocular
  - Kitti dataset only gives labels for one image in the sequence of images
- Instead focusing on loss functions and pre-processing.
  - Quite a breadth of literature on 3D bounding boxes
  - Less publications on occlusion

How Can We Measure Occlusion?

Our Approach: Preprocessing

Approaches

Our Approach: Preprocessing

Our Approach: Preprocessing

Our Approach: Preprocessing

Approaches

Our Approach: Preprocessing

Approaches

Our Approach: Preprocessing

Approaches

Our Approach: Preprocessing

Approaches

Our Approach: Preprocessing

Our Approach: Preprocessing

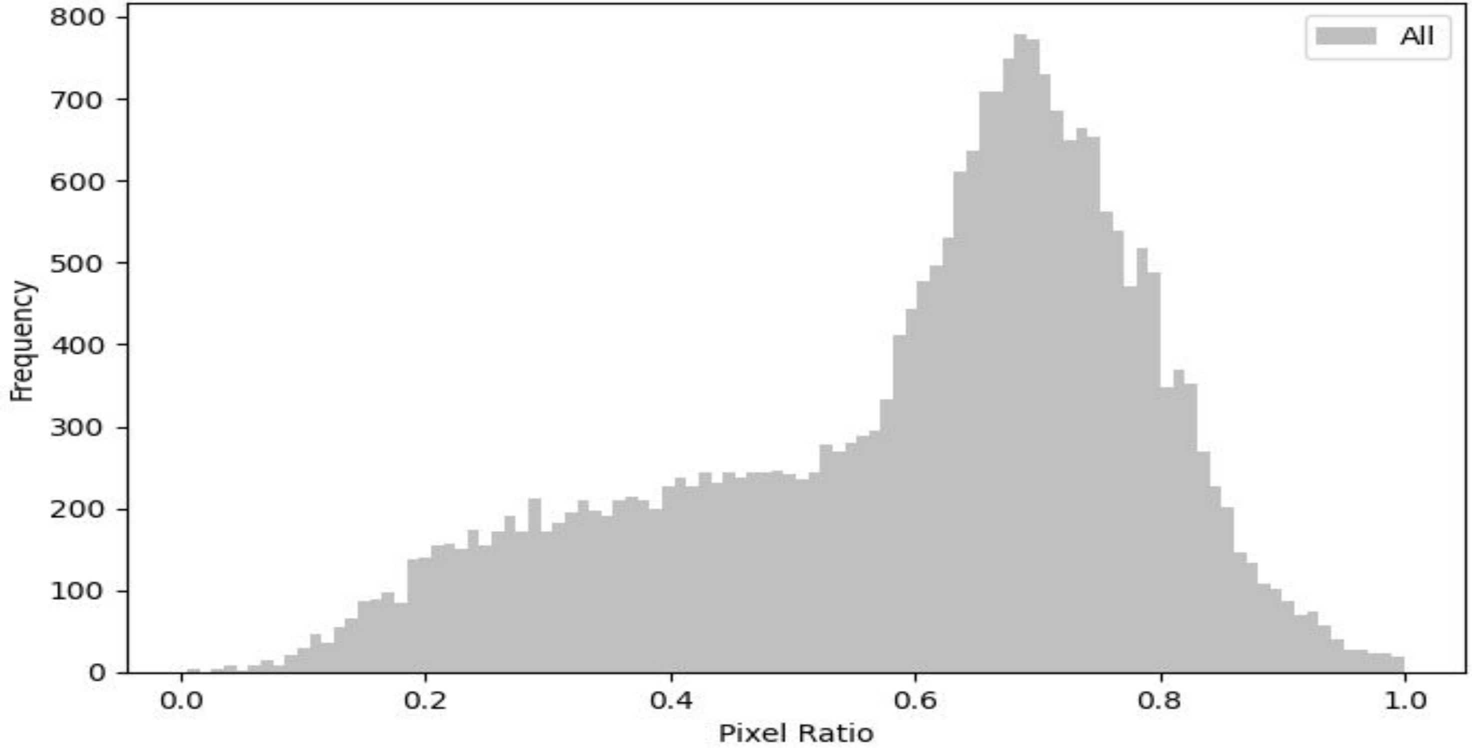
Approaches

Our Approach: Preprocessing

Our Approach: Preprocessing

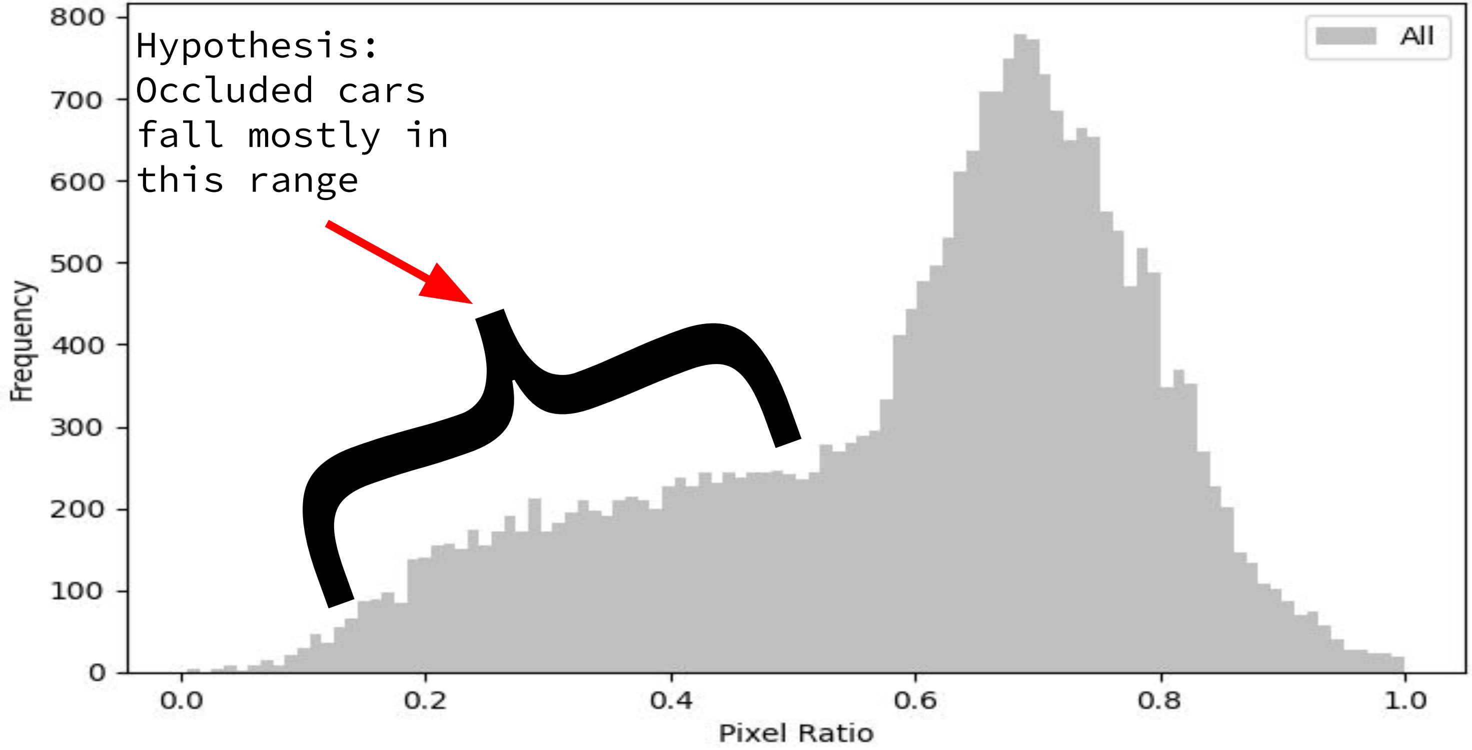
Approaches

Our Approach: Pixel Ratio

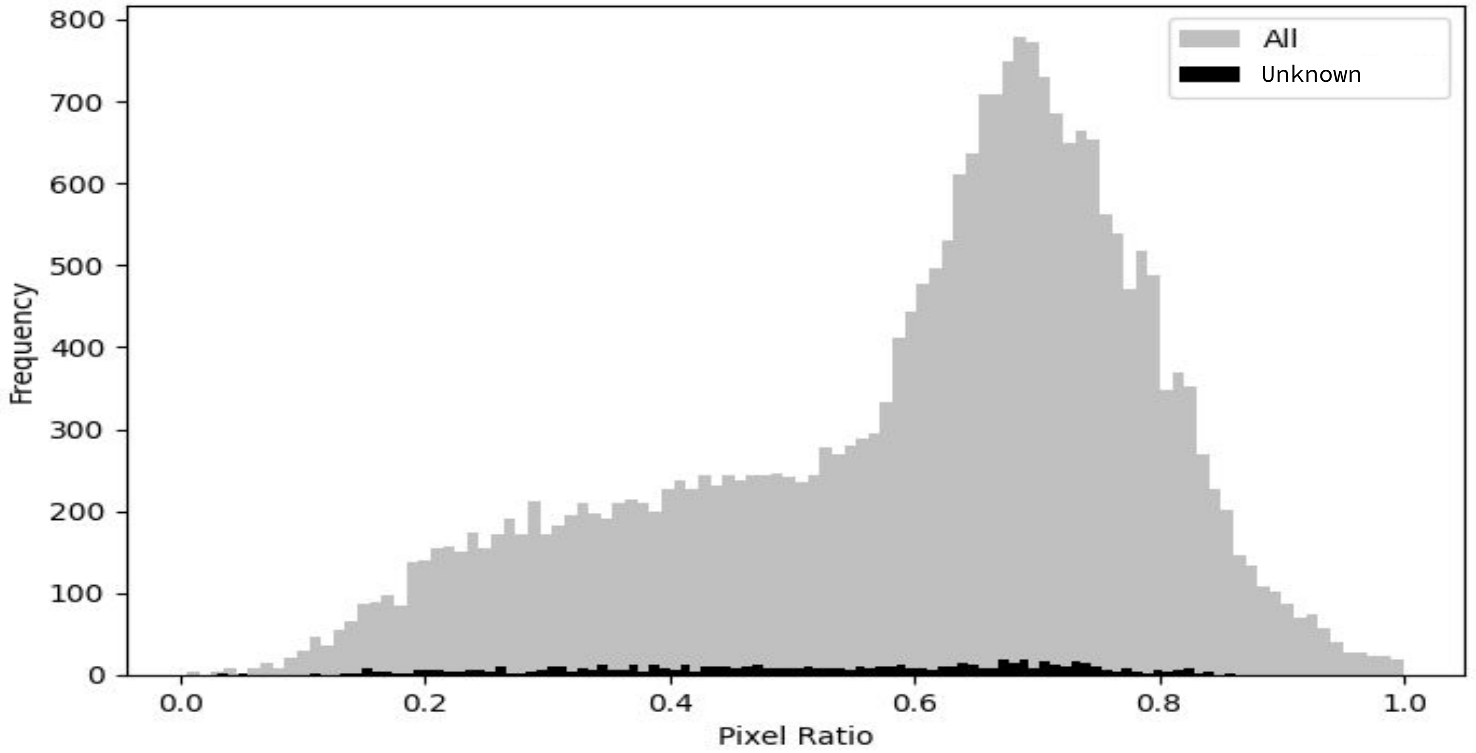


Approaches

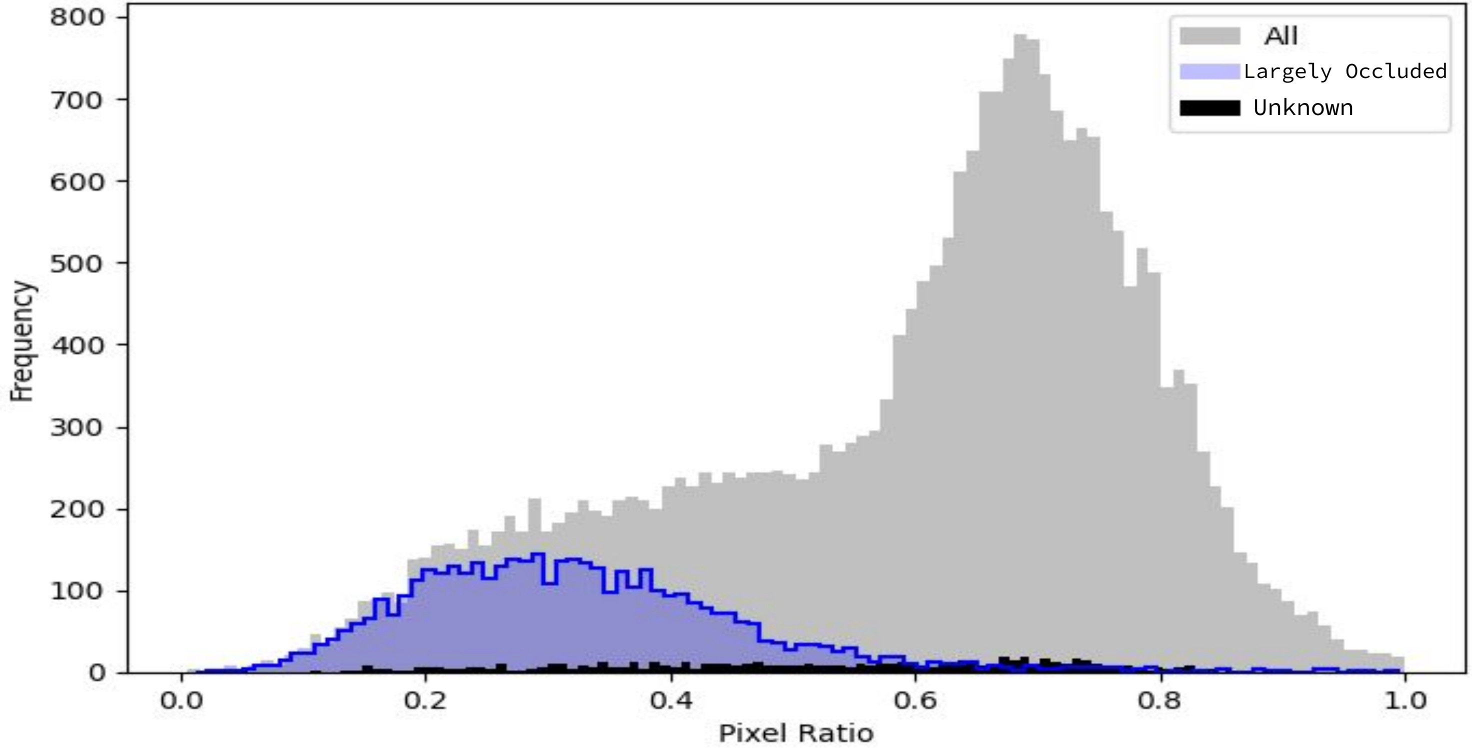
Our Approach: Pixel Ratio



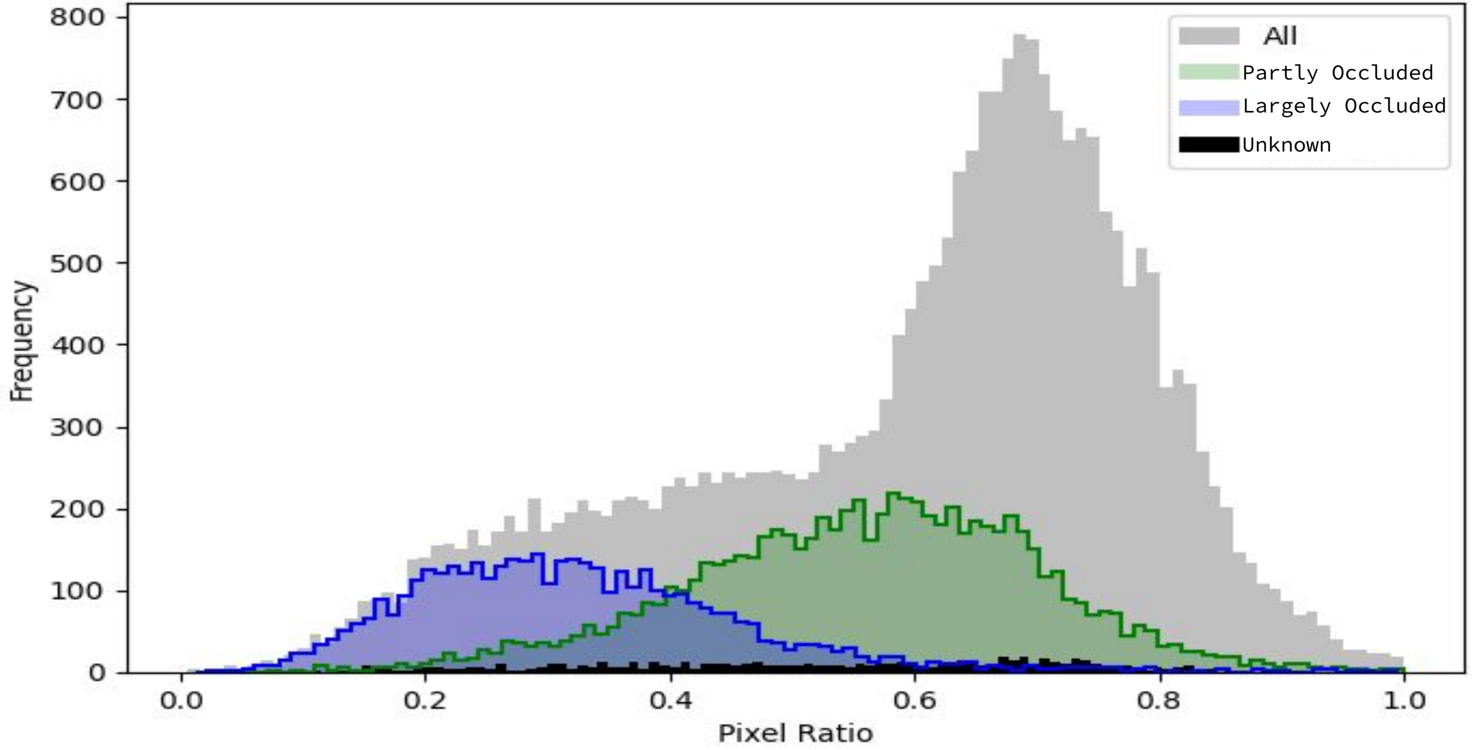
Verified using the KITTI Dataset labels:



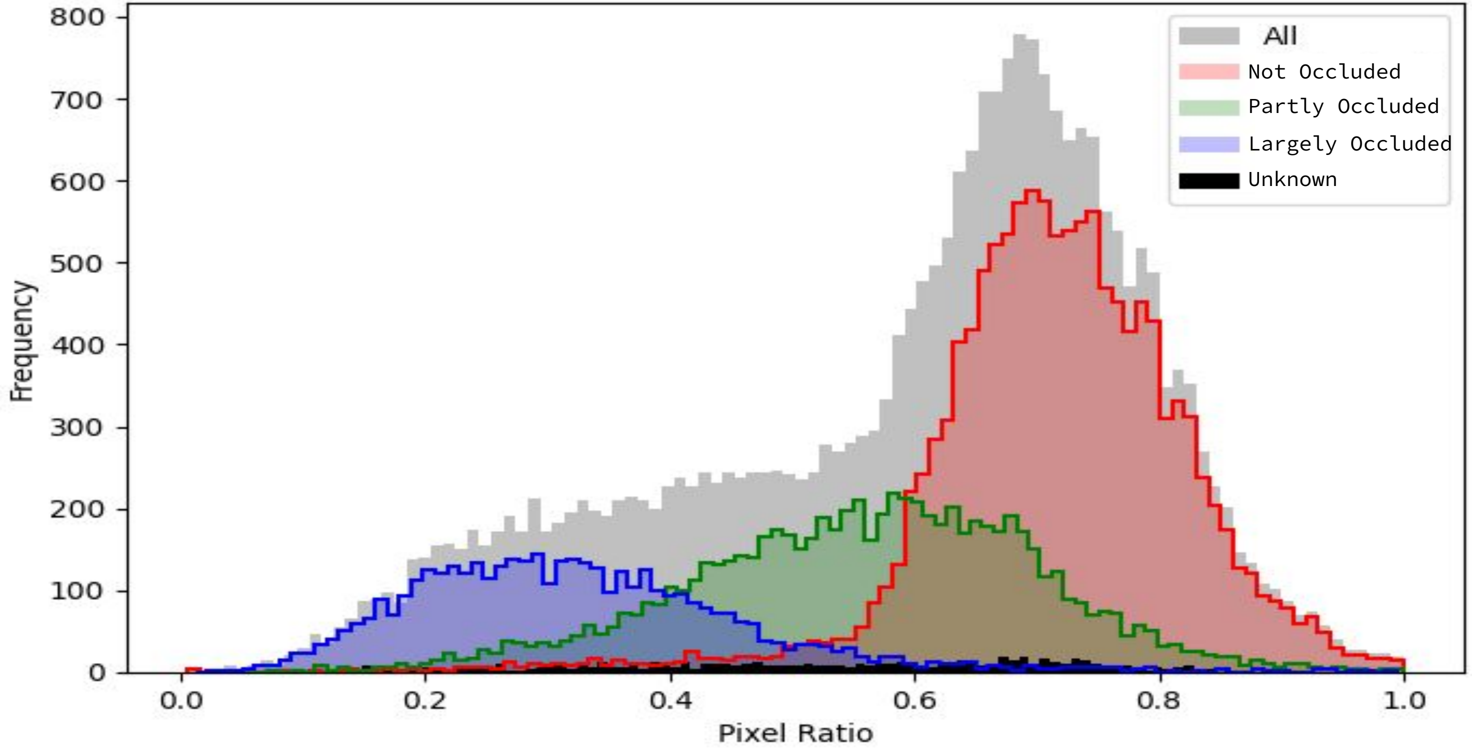
Verified using the KITTI Dataset labels:



Verified using the KITTI Dataset labels:



Verified using the KITTI Dataset labels:



Our Approach: Loss Function

- We are trying several loss functions to optimize for occlusion
- Incorporating pixel ratio in them.

For example:

```
loss = (alpha*  dimension_loss ) +
       (  w  * orientation_loss) +
       (confidence_loss * ratio)
```

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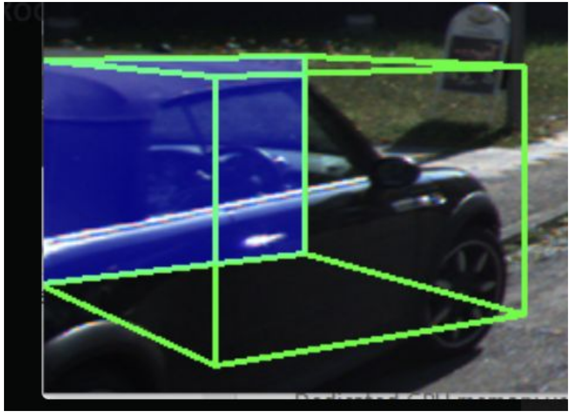
Results

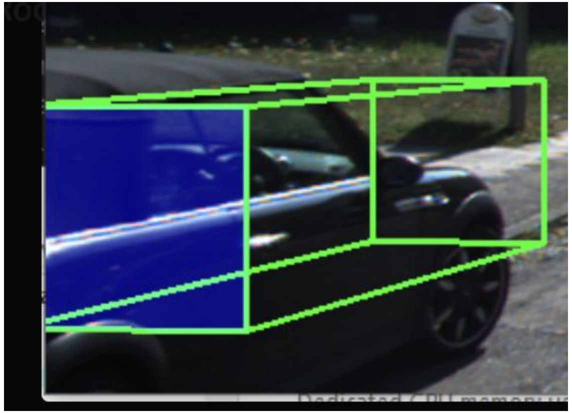
Results

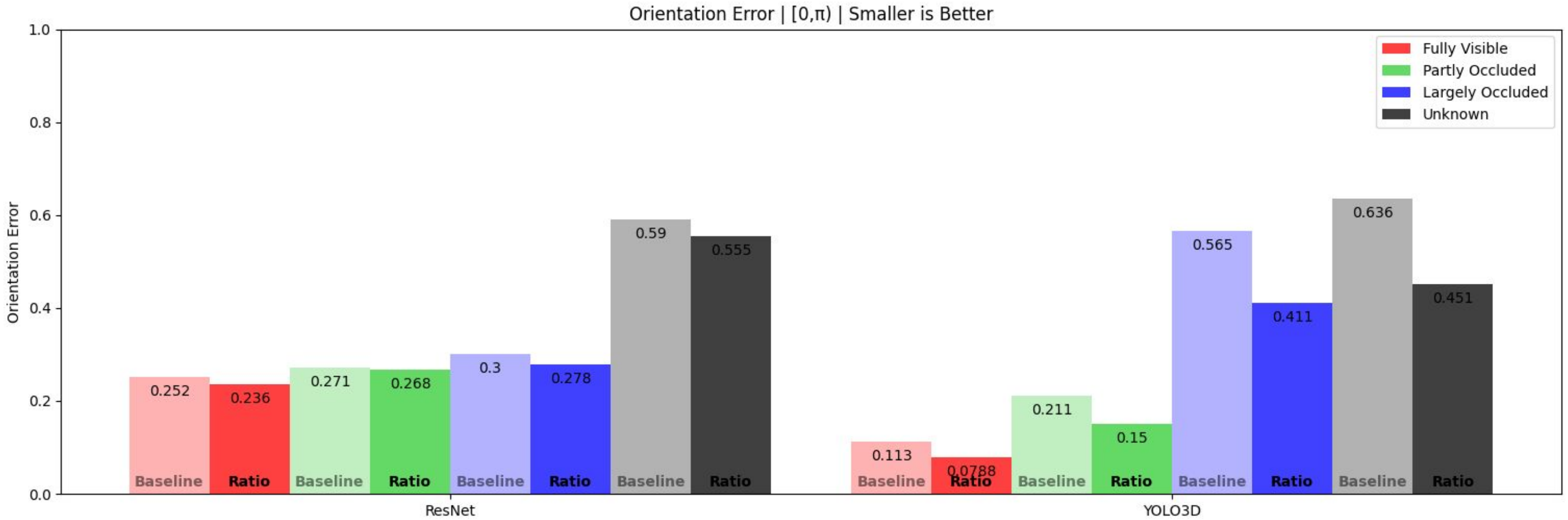
- Our loss function produced better performance on occluded vehicles…

YOLO3D Baseline…



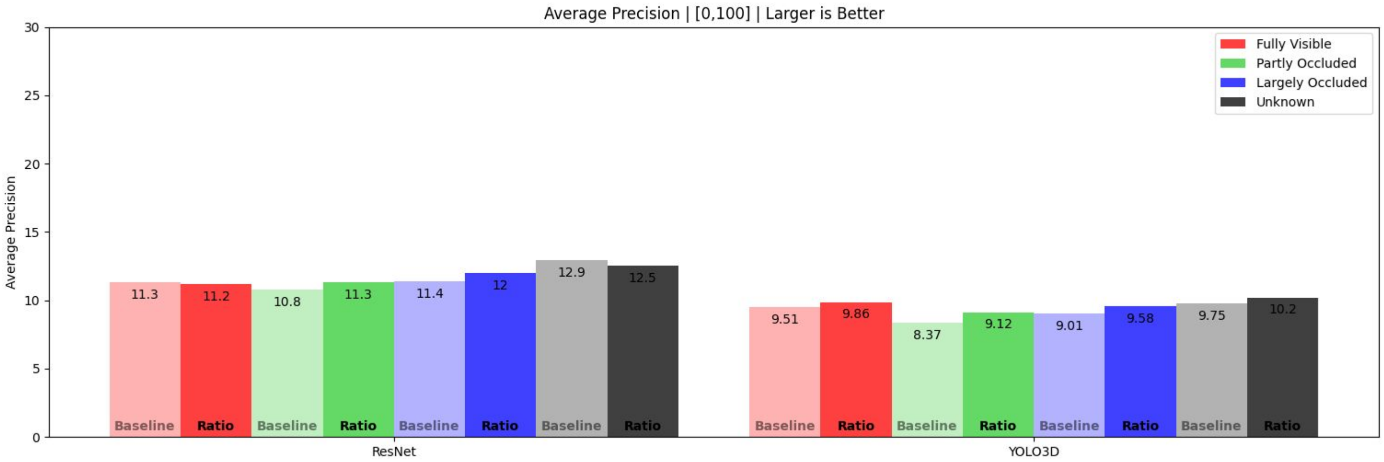
… with our Loss Function

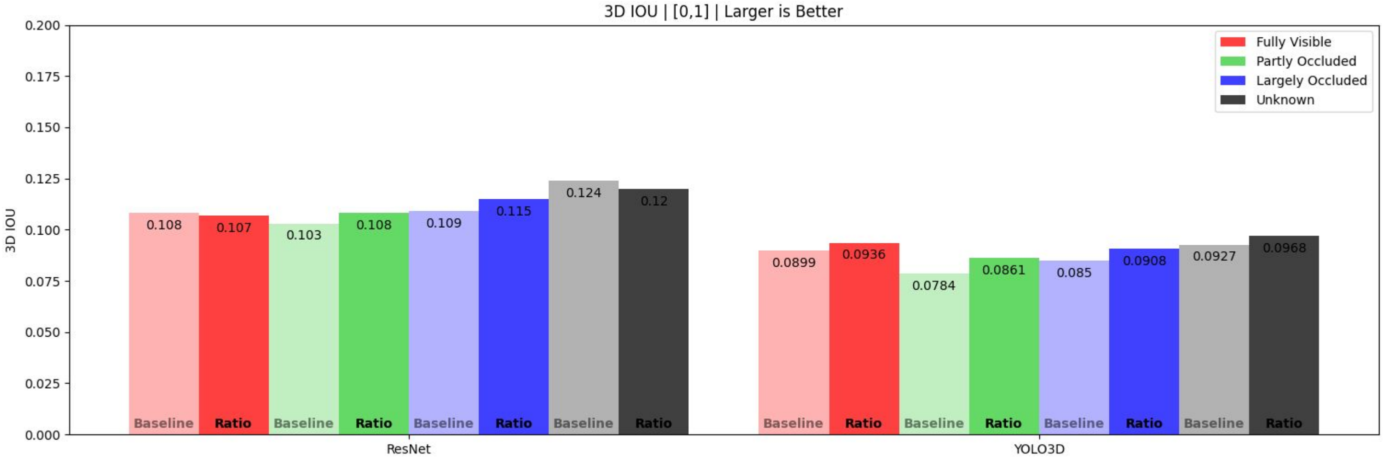




Results

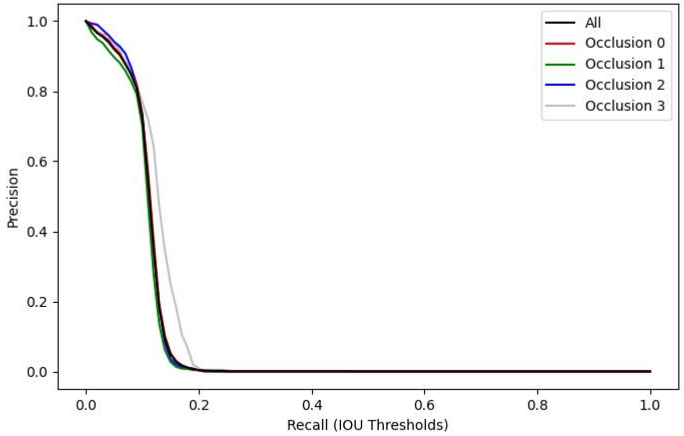
…but only barely



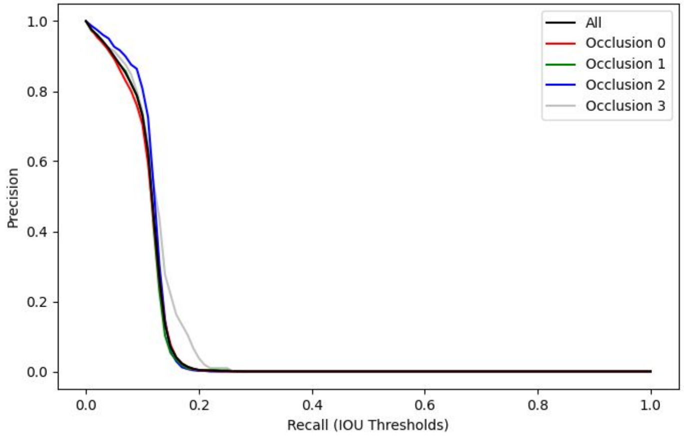


Results

Results



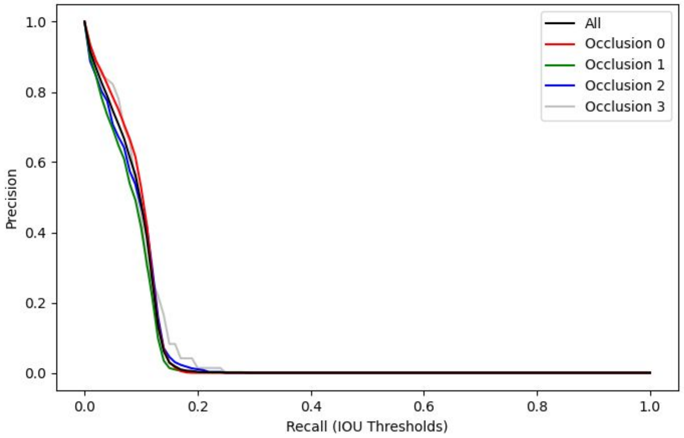
ResNet Baseline
Precision-Recall



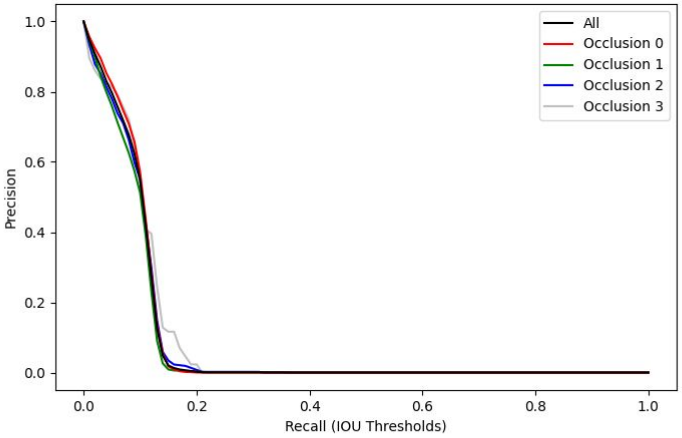
ResNet + Ratio
Precision-Recall

Results

Results



YOLO 3D Baseline
Precision-Recall



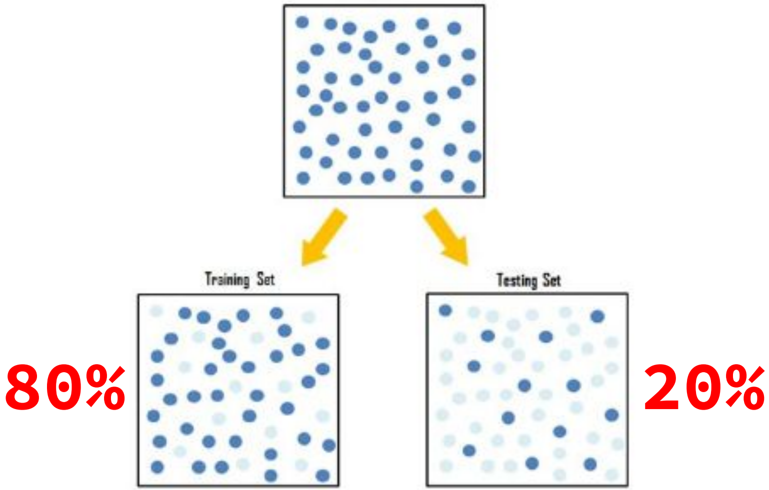
YOLO 3D + Ratio
Precision-Recall

Results

Validation: How is This Being Validated?

- Models trained with variety of methods
  - CV
  - …
- Number of occlusion classes are not equal size!
  - Needed to split by proportion of occlusion class

Train / Test Split!

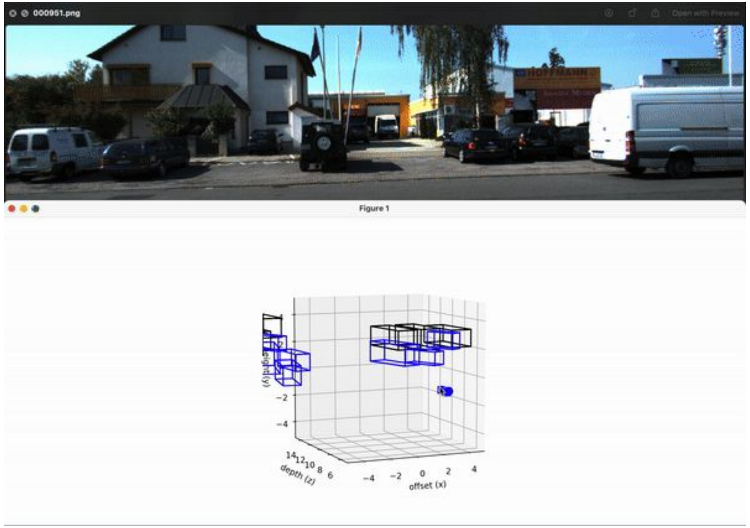


https://ilmudatapy.com/evaluasi-model-machine-learning-dengan-train-test-split/

Results

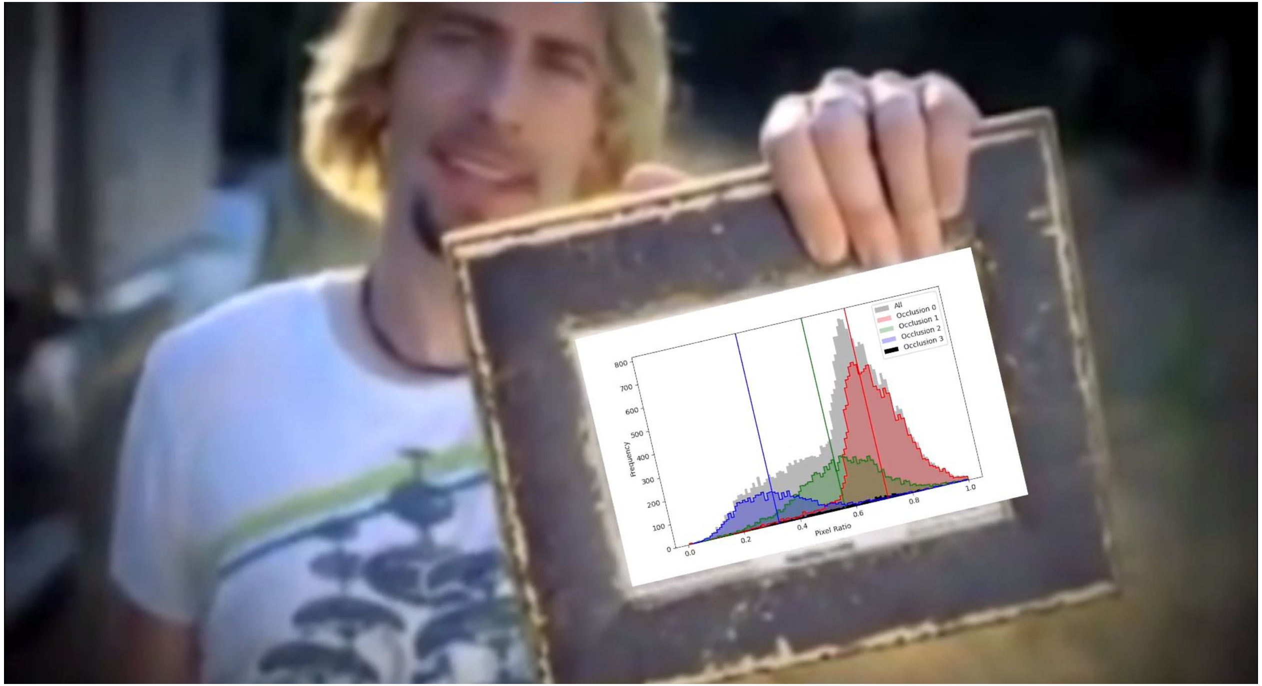
Learnings: What Did We Learn?

- Monocular vision imposes strong limitations

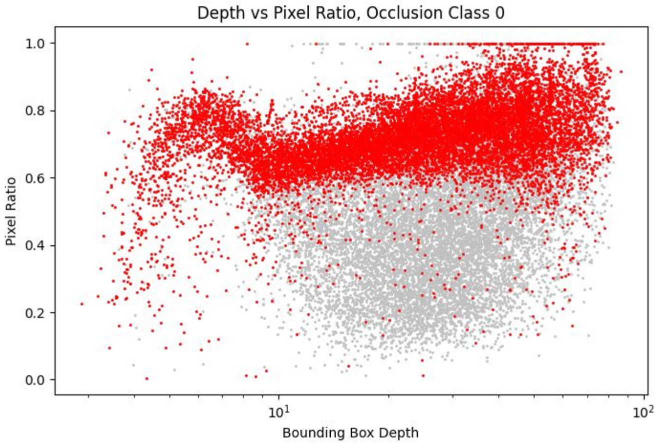


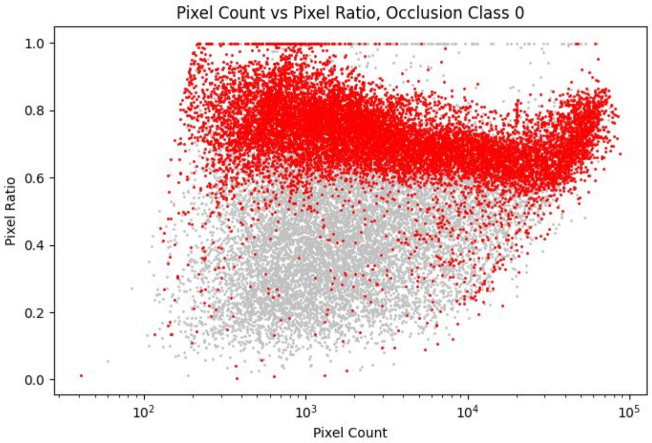
- Computing 3D Rotated IOU is non-trivial
- Improvements are challenging to quantify
    - Is box size or position more important?

Questions?


All
Occlusion 0
Occlusion 1
Occlusion 2
Occlusion 3
Frequency
Pixel Ratio

Is The Pixel Ratio Measure of Occlusion Correlated to Non-Occlusion Variables?

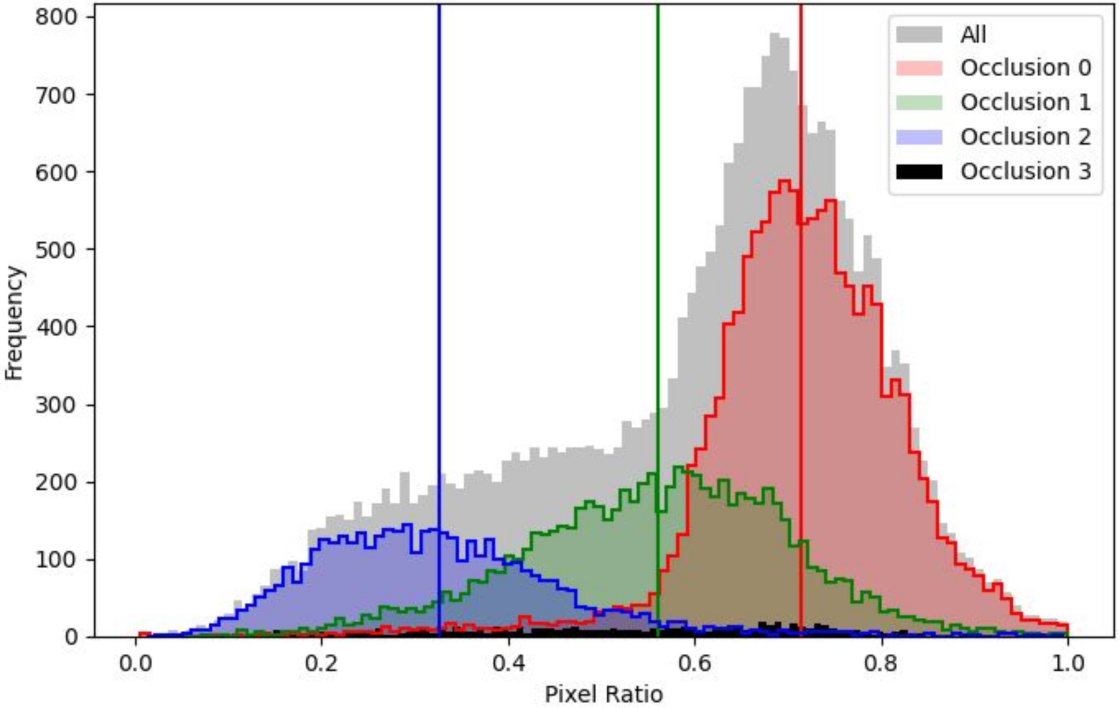




Correlation Coefficient r:

	depth	ratio	pixel_count
depth	1.000000	0.005181	-0.530957
ratio	0.005181	1.000000	0.006493
pixel_count	-0.530957	0.006493	1.000000

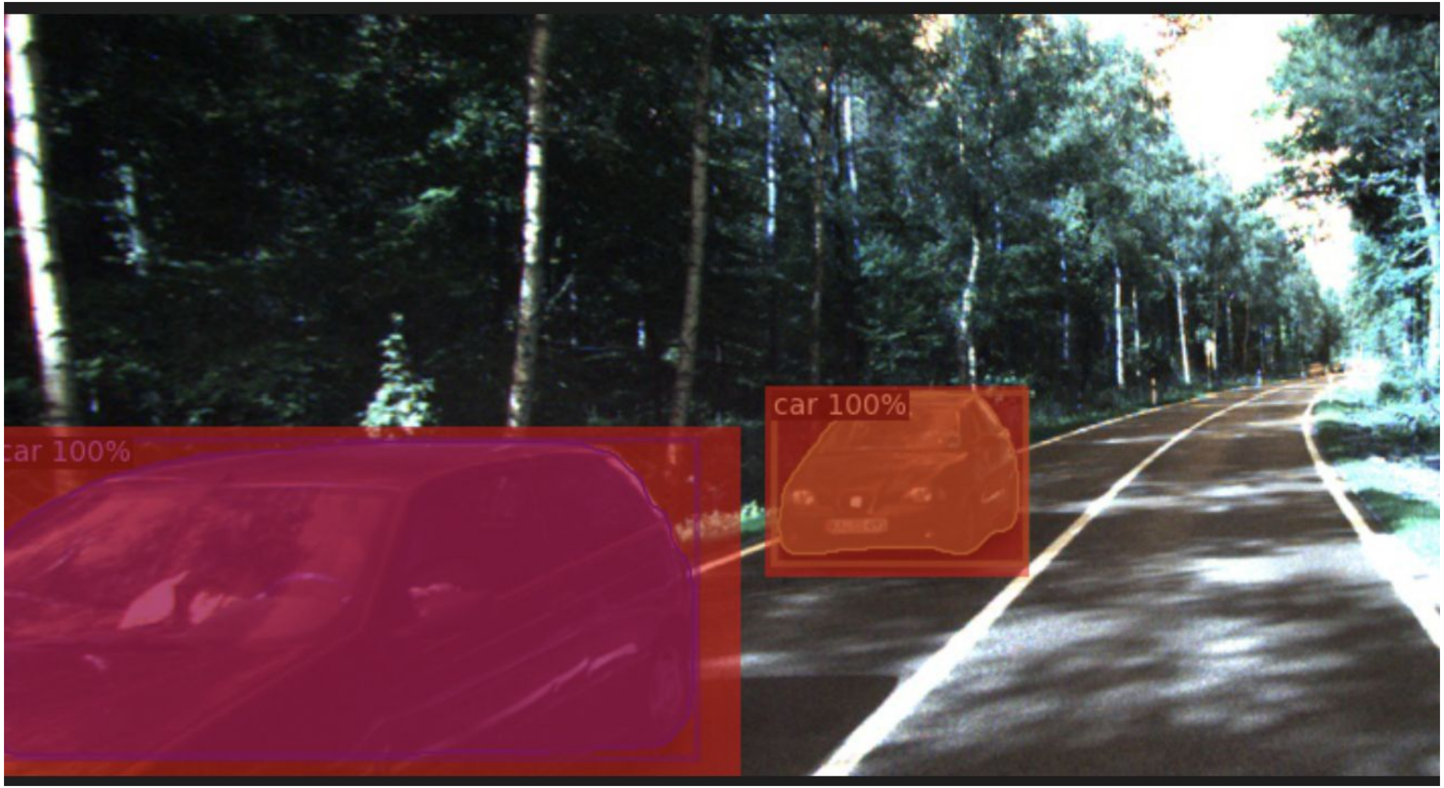
Verified using the KITTI Dataset labels:



ded
uded

Motivation

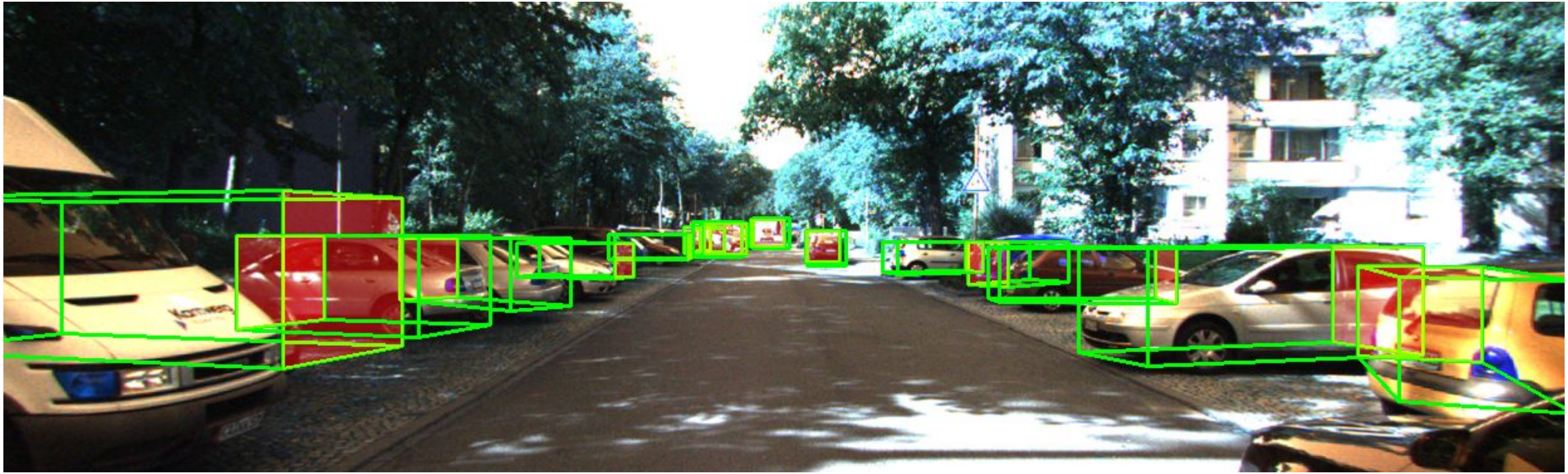
Image bounds occluding car



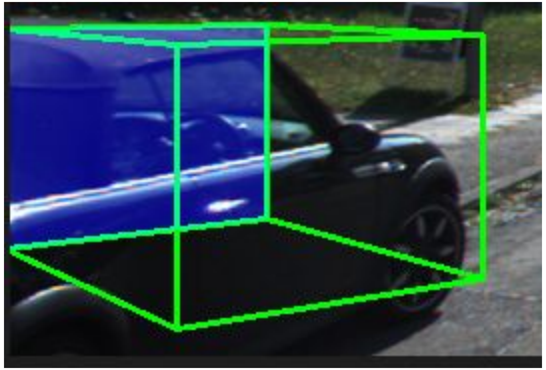
Car occluding car

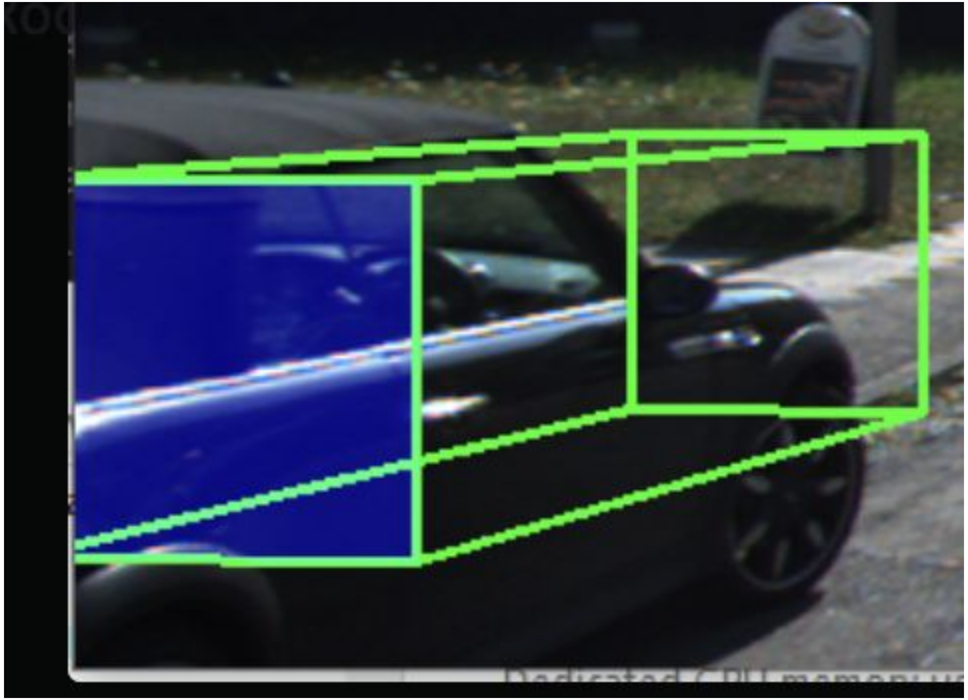
Need to identify both cars as separate entities!


car 100%
car 100%




car 97%
car 89%
car 55%
car 83%



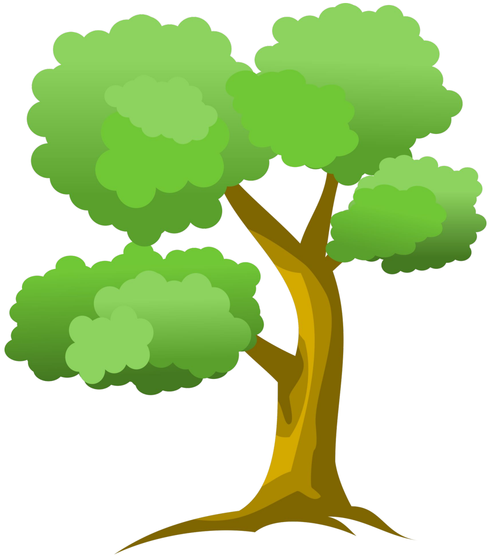



car 100%
car 93%
car 76%
car 92%
motorcycle 62%
car 89%
car 99%
car 98%
car 90%


car 95%
car 100%
car 99%
car 61%
car 80%
car 66%
truck 76%
truck 71%
car 97%
car 57%
car 99%
car 57%


truck 62%
car 100%
car 99%
car 78%
car 100%
car 74%
car 98%
car 96%
car 99%
motorcycle 83%


car 78%
car 95%
car 92%

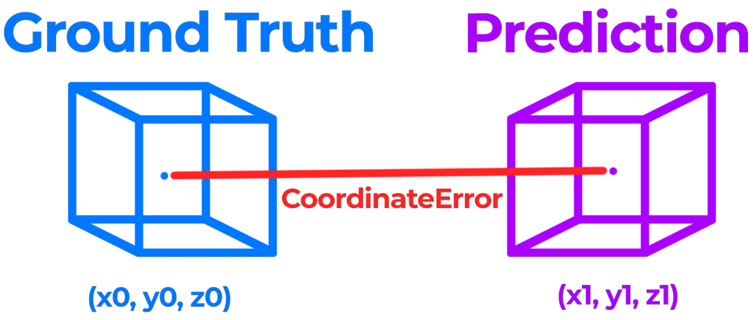


Task Definition

Evaluation: Coordinate Error

- 3D Rotated IOU
- Coordinate Error
- Orientation Error
- Camera Distance Error

- Average Precision



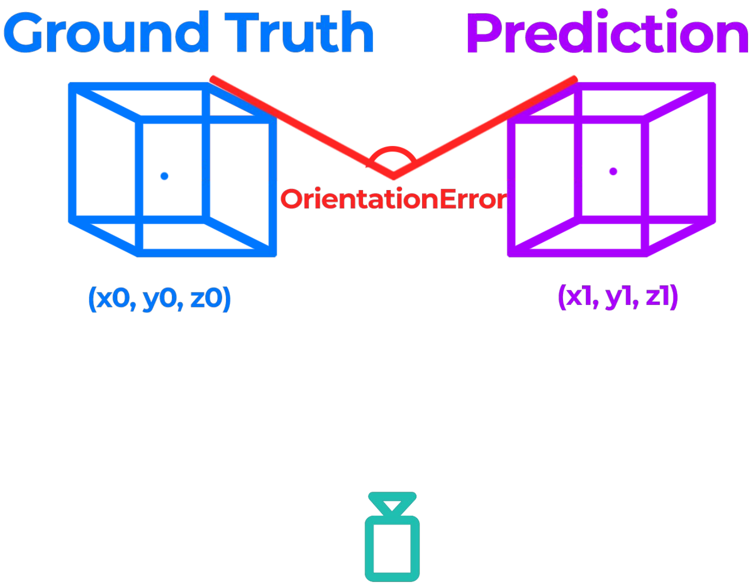
$$\text{CoordinateError} = \sqrt{(x0-x1)^2 + (y0-y1)^2 + (z0-z1)^2}$$

Task Definition

Evaluation: Orientation Error

- 3D Rotated IOU
- Coordinate Error
- Orientation Error
- Camera Distance Error

- Average Precision



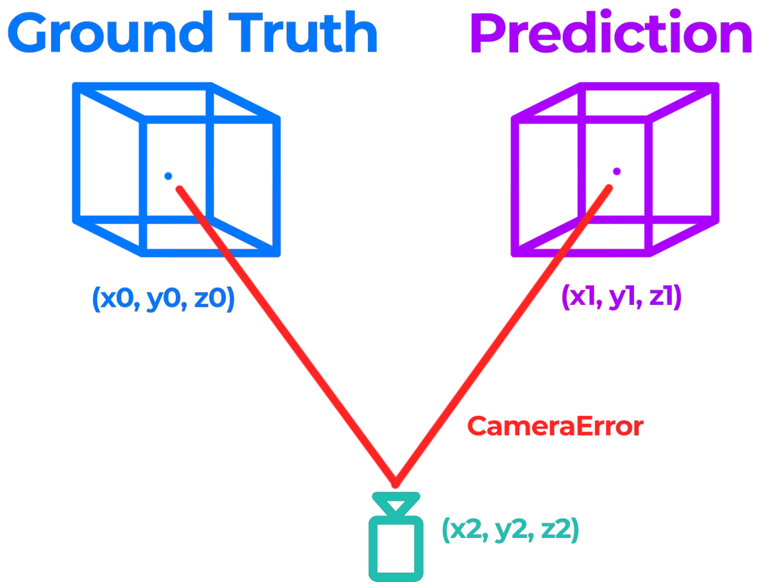
OrientationError = Abs(angle0 - angle1)

Task Definition

Evaluation: Camera Distance Error

- 3D Rotated IOU
- Coordinate Error
- Orientation Error
- Camera Distance Error

- Average Precision



CameraError =

$$\mathrm{abs}\left(\sqrt{(x2-x1)^2+(y2-y1)^2+(z2-z1)^2}-\sqrt{(x0-x2)^2+(y0-y2)^2+(z0-z2)^2}\right)$$

Evaluation: Average Precision

- 3D Rotated IOU
- Coordinate Error
- Orientation Error
- Distance to Camera Error

- Average Precision

1. Choose IOU Threshold

Say IOU > 0.5 is a TP

2. Calculate Precision

Number of bounding boxes processed so far

Precision = TP / (TP + FP)

3. Calculate Recall

Total number of bounding boxes

False Negative?

Recall = TP / (TP + FN)

Task Definition

Evaluation: Average Precision

- 3D Rotated IOU
- Coordinate Error
- Orientation Error
- Distance to Camera Error

- Average Precision

1. Set an IOU value

Recall = IOU!

2. Calculate Precision

Precision = TP / (TP + FP)

Number of bounding boxes processed so far

Task Definition

Evaluation: Average Precision

- 3D Rotated IOU
- Coordinate Error
- Orientation Error
- Distance to Camera Error

- Average Precision

1. **Set an IOU value**

Recall = IOU!

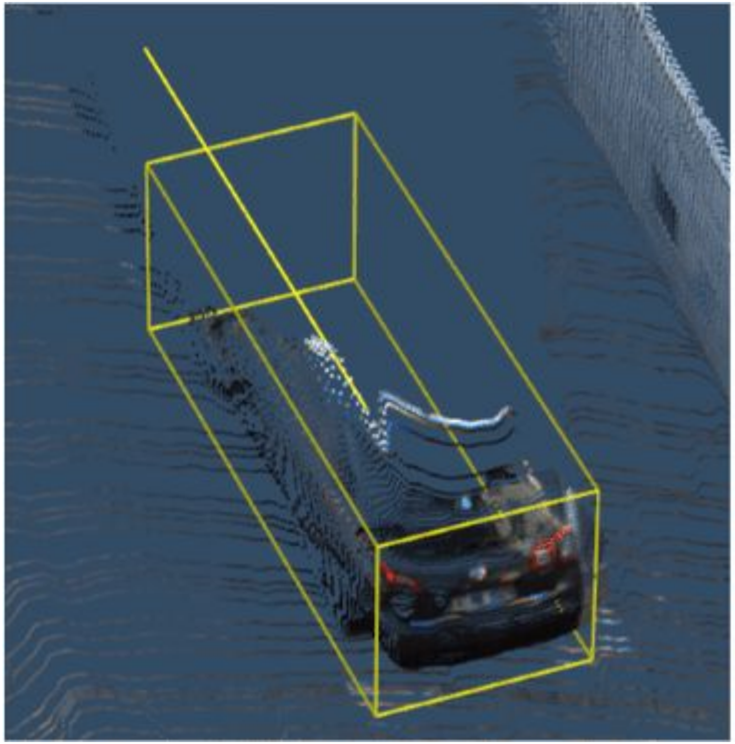
2. **Calculate Precision**

Precision = TP / (TP + FP)

Number of bounding boxes processed so far

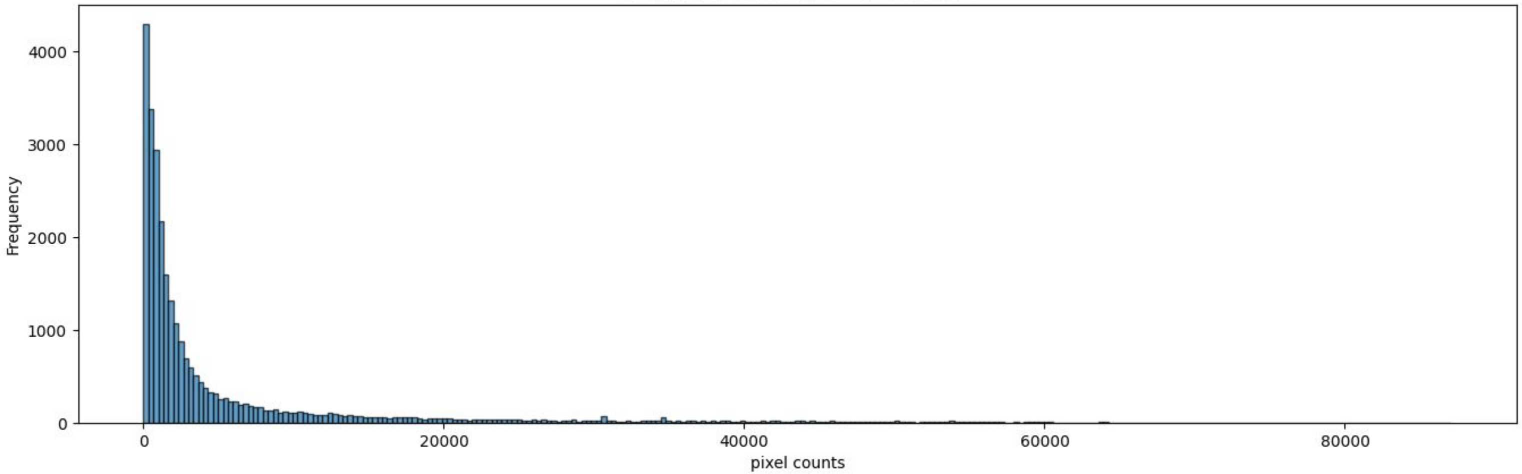
Motivation

Car occluding self *



Approaches

Our Approach: Pixel Ratio



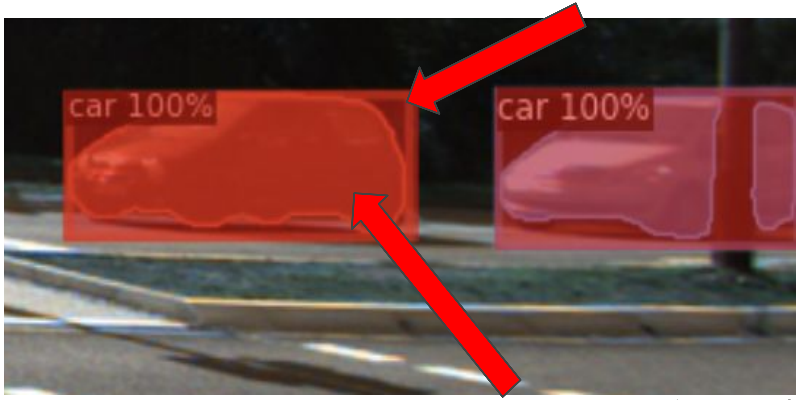
Approaches

Our Approach: Pixel Count

Answering the question:

How can we measure occlusion?

A relationship between the pixel count of an object (obtained using *image segmentation*), and the number of pixels within the 2D bounding box (obtained with *object detection*).



Our Approach: Preprocessing

1. Obtain 2D bounding box of each car from KITTI ground truth

2. Obtain the area of each car in an image using instance segmentation

3. Compute the ratio of (pixel count / 2D bounding box area)

Approaches

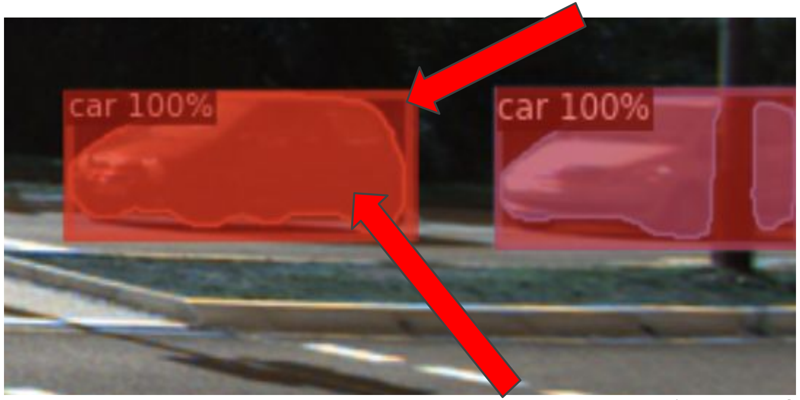
Our Approach: Pixel Count

Answering the question:

How can we measure occlusion?

A relationship between the pixel count of an object (obtained using *image segmentation*), and the number of pixels within the 2D bounding box (obtained with *object detection*).

Bounding Box Area



Pixel count (Area of the car)

Our Approach: Preprocessing

1. Obtain 2D bounding box of each car from KITTI ground truth

2. Obtain the area of each car in an image using instance segmentation

3. Compute the ratio of (pixel count / 2D bounding box area)